

Joshua Varghese

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Minneapolis, MN

Education

University of Wisconsin – Madison

Madison, WI

M.S. in Biomedical Engineering

September 2024 – May 2025

- Coursework: Neuroengineering, Design for Rehabilitation, Tissue Mechanics, Biostatistics, Medical Devices Ecosystem, Biomechanics of Human Movement

B.S. in Biomedical Engineering & Double Major in Data Science

September 2020 – May 2024

- Certificates in Computer Sciences, Statistics
- Coursework: Occupational Ergonomics and Biomechanics, Optimization, Machine Learning, Biochemistry, Data Ethics and Policy, Statistical Experimental Design, Orthopedic Biomechanics, Analytical Chemistry (Honors), Organic Chemistry, Java Programming I-III, Data Science Programming/Modeling I-II

Professional Experience

Hardware Engineer II

St. Paul, MN

Abbott Laboratories

May 2025 - Present

- Designed and implemented Python-based automated test frameworks interfacing with microcontrollers, embedded firmware, and precision measurement equipment to accelerate product-level test execution and improve overall reliability.
- Authored, executed, and documented Design Verification and Validation protocols for the Volt Pulsed Field Ablation and TactiFlex Duo RF/PF systems, ensuring traceability to firmware, product, and system level requirements.
- Identified, diagnosed, and characterized defects through structured experimentation and data-driven analysis, partnering with cross-functional teams to define root causes and implement corrective actions.
- Provided hands-on laboratory support during GLP studies and LMR investigations by reproducing field issues, evaluating system behavior, and delivering rapid feedback to clinical, quality, and engineering stakeholders.
- Ensured adherence to Good Documentation Practices (GDP) and design controls through rigorous document management in Windchill and issue tracking, traceability, and closure using Jira.

Bioinstrumentation (BME 310) Teaching Assistant

Madison, WI

UW-Madison Department of Biomedical Engineering

January 2025 – May 2025

- Designed and tested medical device sensor circuits for labs (e.g. pressure, ECG, temperature, ultrasonic).
- Troubleshoot analog circuits using oscilloscopes and DMMs, supporting student designs.
- Reinforced students' understanding of signal-processing fundamentals by instructing them in the development of operational amplifier circuits and filter circuits.
- Led two lab sections and hosted office hours while collaborating with faculty to enhance course materials.

R&D Translational Medicine Scientist

Pearl River, NY

Pfizer Inc.

May 2024 - February 2025

- Independently developed, trained, and tested an AI tool using spaCy, NLTK, and PyTorch to automate literary data extraction for systematic reviews, successfully reducing analysis time by 57%.
- Analyzed clinical trial data and collaborated with regulatory teams to address feedback, ensuring compliance and significantly improving study quality.
- Designed, built, and deployed a user-friendly data analysis application using Posit Connect and Snowflake, independently training end-users and ensuring successful adoption across teams.
- Conducted a systematic review on predictive, non-invasive IBD biomarkers that contributed to a manuscript summarizing treatment options and advancements in the field.

Undergraduate Researcher / Bioinformatician

Madison, WI

Aids Vaccine Research Lab

May 2021 - May 2024

- Developed an active noise-canceling system with Raspberry Pi, reducing motor and fan noise from large air samplers.
- Prepared and sequenced cynomolgus macaque DNA on Oxford Nanopore Technology platforms (GridION and MinION), using Bash and Shell commands on Mac and Unix systems for data analysis and visualization.
- Deployed reproducible projects on high-performance computing servers at Wisconsin's Center for High Throughput Computing and developed custom Geneious Prime GUI plugins for sequence analysis, ensuring efficient workflows.
- Automated the upload of 125,000+ COVID sequences to public repositories using Selenium, improving efficiency by 300% and streamlining data submission processes.
- Led recruitment, training, and supervision of undergraduate researchers, ensuring effective team development.

- Designed and implemented a 3D model for hybrid air cartridges to improve accessibility for low-income groups.

Global Technology and Engineering Biotech Data Visualization Intern

Sanford, NC

Pfizer Inc.

May 2023 - August 2023

- Analyzed and optimized process workflows within the Manufacturing Sciences and Technology department, enhancing large-scale manufacturing efficiency and ensuring compliance with cGMP SOPs.
- Leveraged a time-series database and Visual Basic to extract and query experimental run data from the Aveva PI System, streamlining data retrieval for process improvements.
- Engineered and took ownership of a PowerBI dashboard to generate dynamic visuals and statistical tables for bio-suitability reports, reducing presentation time by 80%.
- Automated qPCR data analysis in a SQL server environment, applying visualizations and statistical methods to validate standard curves and assay acceptances, driving data-informed decision-making.
- Developed and implemented robust data backup procedures and led knowledge transfers, ensuring operational continuity and smooth team transitions.

Engineering Design Experience

Reducing Whole-Body Vibrations in Neonatal Transport Team Leader

Madison, WI

UW-Madison School of Medicine and Public Health, Department of Neonatology

August 2022 - May 2024

- Led a multidisciplinary team to design and prototype solutions to reduce neonatal transport vibrations.
- Conducted comprehensive literature reviews, modeled components in SolidWorks, and fabricated prototypes using 3D printing and machining, achieving a 50% reduction in vibrations experienced by neonates.
- Developed and executed test plans in mechanical engineering labs and ambulances, presenting findings to academic and industry stakeholders.

Safer Emesis Container Biomedical Student Advising Committee Member

Madison, WI

UW-Madison College of Engineering

August 2021 - December 2021

- Built a prototype capable of capturing and containing Norovirus particles expelled during emesis.
- Conducted a literature review, designed and modeled prototypes in SolidWorks, fabricated the final design, and presented findings in a formal poster presentation.
- Simulated particle flow using Ansys to optimize design efficiency.
- Provided insights on structuring the design course for all Biomedical Engineering students at UW-Madison.

Patents and Publications

- Hassanein, M., Xi, L., Ryan, R., Ye, Z., Tania, N., Sano, Y., Varghese, J., et. al. Advancing IBD Management: A Literature Review on the Role of Non-Invasive Blood-Based Biomarkers in Predicting and Assessing Pharmacodynamic Response to Treatment. *Clinical and Translational Science*.
- Karl, J., Prall, T. M., Bussan, H. E., Varghese, J. M., et. al. Complete sequencing of a cynomolgus macaque major histocompatibility complex haplotype (p. 2022.10.24.513544). *bioRxiv*.
- Prall, T. M., Karl, J. A., Varghese, J. M., et. al. Complete genomic assembly of Mauritian cynomolgus macaque killer immunoglobulin-like receptor and natural killer group 2 haplotypes. *Genomics*.
- NeoSafe Vibration Shield. P250035US01, filed December 6, 2024. Patent pending.

Technical Skills

- **Programming & Scripting:** Python, R, Java, SQL, MATLAB, HTML, Shell, Visual Basic, Arduino
- **Software Tools:** Git, Jenkins, Docker, Perforce, LTSpice, Altium
- **Product Lifecycle & Project Management:** Windchill, Jira
- **Data Analysis & Visualization:** Tableau, Power BI, Microsoft Excel
- **Productivity & Design:** Microsoft Office Suite (Word, PowerPoint, Outlook, Visio), Adobe Creative Cloud (Photoshop, Illustrator, Acrobat)